

Variational Quantum Monte Carlo of Hydrogen Molecule

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Abstract—This project used Variational Monte Carlo (VMC) to estimate the ground-state energy and equilibrium bond length of the hydrogen molecule H_2 and to quantify the deviation of the calculated results from accepted experimental values. The numerical components of the method were first validated on systems with known analytical solutions, including tests of the central-difference Laplacian, Metropolis sampling, and simulated-annealing parameter optimisation in both one and three dimensions. The full VMC method was then applied to the two-electron hydrogen molecule using a Slater–Jastrow trial wavefunction. Variational parameters were optimised using a vectorised multi-walker Metropolis scheme with simulated annealing. The resulting energy–bond-length curve was fitted to a Morse potential, giving an equilibrium separation $r_0 = 1.3223$ a.u. and a minimum energy $E_{\min} = -1.1565$ a.u., both within approximately 10% of accepted physical values. These results demonstrate that a VMC implementation can provide accurate predictions for molecular systems.

I. INTRODUCTION

THE aim of this numerical experiment was to use Variational Monte Carlo (VMC) to estimate the ground-state energy of H_2 and to quantify the deviation of the calculated results from accepted experimental values.

Calculating the ground-state energy of a molecule is essential in quantum physics, as it determines whether the molecule is bound and therefore whether it can exist at all. The time-independent Schrödinger equation (TISE) defines the allowed energy levels of a quantum system [1]. However, when more than one electron is present, the Coulomb interaction between electrons makes the equation non-separable, and no analytical solution exists [1].

The hydrogen molecule is the simplest system that exhibits this difficulty. It therefore serves as an ideal minimal test case for studying the effectiveness of numerical methods.

Direct numerical approaches, such as discretising configuration space, are also infeasible. For N particles in d dimensions, the wavefunction depends on

Nd variables, $\psi(r_1, r_2, \dots, r_N)$. Storing this function on a grid requires memory that scales exponentially with the number of degrees of freedom. For example, using only 100 points along each coordinate and storing each value as a 64-bit float would require 64×100^{Nd} bits. This exponential increase makes grid-based solutions impractical for higher dimensions. These challenges motivate the use of methods that rely on random sampling such as VMC.

II. THEORY

Monte Carlo Integration

The ground state of a quantum system is the state that minimises the expectation value of the Hamiltonian. For a system of N electrons in three dimensions with trial wavefunction $\psi(r_1, \dots, r_N)$, the expectation value of the Hamiltonian is

$$\langle H \rangle = \frac{\int d^3r_1 \dots d^3r_N \psi^*(r_1, \dots, r_N) \hat{H} \psi(r_1, \dots, r_N)}{\int d^3r_1 \dots d^3r_N |\psi(r_1, \dots, r_N)|^2}. \quad (1)$$

Writing $R = (r_1, \dots, r_N)$ and assuming a real wavefunction, we can substitute the local energy,

$$E_l(R) = \frac{\hat{H} \psi(R)}{\psi(R)}, \quad (2)$$

and the normalised probability density,

$$\rho(R) = \frac{|\psi(R)|^2}{\int dR |\psi(R)|^2}, \quad (3)$$

to obtain

$$\langle H \rangle = \int dR \rho(R) E_l(R). \quad (4)$$

In Monte Carlo integration, we approximate this expectation value by sampling configurations $R^{(i)}$

from $\rho(R)$ and computing the sample mean of the local energy:

$$\langle H \rangle \approx \frac{1}{N_s} \sum_{i=1}^{N_s} E_l(R^{(i)}), \quad (5)$$

where N_s is the number of samples.

Variational Monte Carlo

VMC is a numerical method for estimating the ground-state energy of a quantum system using the variational principle together with random sampling [2]. The variational principle states that $\langle H \rangle \geq E_0$ [1], where E_0 is the exact ground-state energy. The method begins with a trial wavefunction $\psi(R; \theta)$, where R represents the full configuration of all electrons and θ is a set of variational parameters. By minimising $\langle H \rangle$ with respect to θ , we obtain the best estimate of the ground-state energy that this wavefunction form can provide.

The variational energy is computed by generating random samples of the configuration R distributed according to $|\psi(R)|^2$. Once a representative set of configurations has been obtained, the ground-state energy estimate is computed by averaging the local energy over these samples.

III. METHODS AND RESULTS

A. Numerical Derivative Accuracy Test

Method: To ensure that the numerical Laplacian used later in the VMC calculations is reliable, I first tested the accuracy of the second-derivative operator used throughout the project. The aim was to identify a stable range of step sizes h for which the finite-difference Laplacian is accurate and not dominated by round-off error. The optimal choice of h is set primarily by machine precision rather than the details of the wavefunction, so a single value can be used consistently in later three-dimensional calculations.

The test function was the ground-state harmonic-oscillator wavefunction, which satisfies the TISE

$$-\frac{\hbar^2}{2m_e} \frac{d^2\psi}{dx^2} + \frac{1}{2}m_e\omega^2 x^2 \psi(x) = E \psi(x), \quad (6)$$

and becomes, after nondimensionalising with $\hat{x} = x/L$ and $L = \sqrt{\hbar/(m_e\omega)}$,

$$H = -\frac{1}{2} \frac{d^2}{d\hat{x}^2} + \frac{1}{2} \hat{x}^2. \quad (7)$$

Its analytical eigenfunctions and eigenvalues are

$$\psi_n(x) = H_n(x) e^{-x^2/2}, \quad E_n = n + \frac{1}{2}, \quad (8)$$

For the ground state $n = 0$, for which $H_0(x) = 1$ and therefore both $\psi(x)$ and its exact second derivative $\psi''(x)$ are known in closed form.

The numerical Laplacian was approximated using the 2nd-order central-difference scheme (CDS)

$$\psi''_{\text{num}}(x; h) = \frac{\psi(x+h) - 2\psi(x) + \psi(x-h)}{h^2}, \quad (9)$$

which has truncation error of order $\mathcal{O}(h^2)$.

To assess its accuracy, I evaluated $\psi''_{\text{num}}(x_0; h)$ at a fixed point x_0 over a range of step sizes. The absolute error was computed for each h . I also repeated the test using the 4th-order CDS to determine whether its higher-order accuracy made it preferable for the later three-dimensional VMC calculations.

Results: Figure 1 shows the absolute error in the numerical second derivative as a function of the step size h for both the 2nd-order and 4th-order CDS. The results confirm the expected truncation-error behaviour predicted by Taylor-series analysis.

For large step sizes ($2 \times 10^{-4} \lesssim h$), the error of the 2nd-order CDS follows an h^2 scaling. For the 4th-order CDS ($4 \times 10^{-3} \lesssim h$), the error follows an h^4 scaling. These trends agree with the reference slope lines in the figure, demonstrating that both finite-difference formulas achieve their formal order of accuracy.

At smaller step sizes ($h \lesssim 2 \times 10^{-4}$ for the 2nd-order CDS, and $h \lesssim 4 \times 10^{-3}$ for the 4th-order CDS), the error no longer decreases and instead begins to rise. This deviation indicates the onset of floating-point round-off error: as h becomes too small, the numerator in the finite-difference expression involves subtracting nearly equal quantities, which reduces numerical precision.

The minimum error within the region not dominated by round-off was of order 10^{-9} for the 2nd-order central difference scheme, and of order 10^{-11} for the 4th-order scheme. Despite the 4th-order CDS achieving a smaller minimum error, I chose to use the 2nd-order CDS for all subsequent numerical derivatives at the optimal step size $h = 2 \times 10^{-4}$ because its derivative error (of order 10^{-9}) is already negligible compared with the Monte Carlo statistical noise in the final energy estimates, so the additional

cost of the 4th-order stencil would not yield any practical improvement in accuracy.

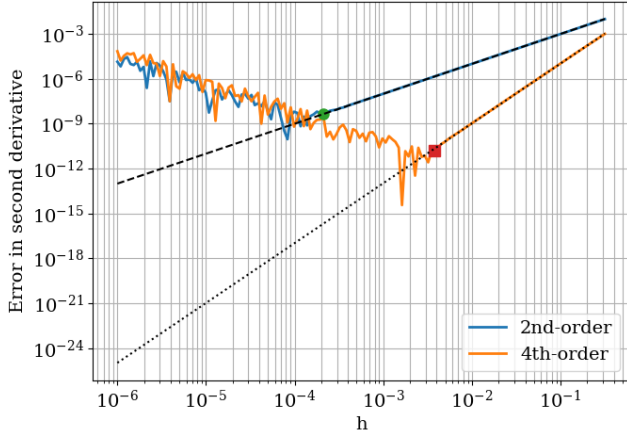


Fig. 1. Finite-difference error scaling for the numerical second derivative using 2nd-order and 4th-order stencils (log-log axes). The dotted and dashed lines show the expected h^2 and h^4 slopes. The green circle and red square indicate the minimum-error points for the 2nd- and 4th-order stencils, respectively.

B. Metropolis Algorithm Estimation in 1D

Method: I applied the Metropolis algorithm to the one-dimensional harmonic oscillator in order to verify that the sampling procedure was working correctly before extending it to three dimensions.

Solving Eq. (5) yields the probability density functions. For the ground state ($n = 0$):

$$\rho_0(x) = \pi^{-1/2} e^{-x^2}. \quad (10)$$

For the first excited state ($n = 1$):

$$\rho_1(x) = 4x^2 \pi^{-1/2} e^{-x^2}. \quad (11)$$

The Metropolis update was implemented by proposing a random step $x' = x + \Delta$. The Metropolis proposal displacement Δ was drawn uniformly from the interval $[-\Delta_{\max}, \Delta_{\max}]$. I tested several values of $\Delta_{\max}$ and selected the one that produced an acceptance rate of approximately 50%. I found that after roughly 10^6 iterations the sampled distribution no longer changed noticeably. Further iterations did not improve the distribution but only increased runtime, so I adopted 10^6 iterations as a practical and sufficient choice.

Each proposed move was accepted with probability [2]

$$\alpha = \min\left(1, \frac{\rho(x')}{\rho(x)}\right). \quad (12)$$

After 10^6 iterations, I estimated the variational energy by averaging the local energy over the sampled positions as defined in Eq. (5).

Results: Figure 2 shows the histogram of samples produced by the Metropolis algorithm for the one-dimensional harmonic oscillator. The sampled distribution aligns closely with the analytical density $\rho(x) = \pi^{-1/2} e^{-x^2}$, indicating that the Markov chain successfully converged to the correct equilibrium distribution.

I obtained energy values of $E = 0.4999997$ for the ground state and $E = 1.4999955$ for the first excited state, both in excellent agreement with the exact values of 0.5 and 1.5. These agree with the exact energies to within 10^{-6} , confirming the correctness of the sampling procedure and the numerical evaluation of the local energy.

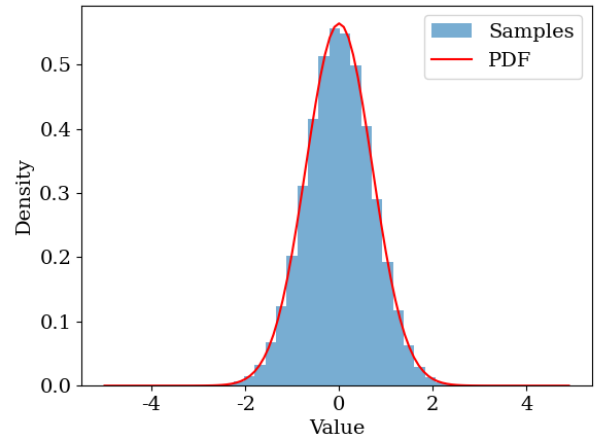


Fig. 2. Histogram of Metropolis-Hastings samples for the 1D harmonic oscillator compared with the exact probability density.

C. Extension to 3D: Ground State of Hydrogen

Method: I then applied the Metropolis sampler from the one-dimensional case together with simulated annealing[3] to implement the full VMC method for the hydrogen atom in 3D. I used a simple hydrogen-like trial wavefunction of the form

$$\psi(\mathbf{r}; \theta) = e^{-\theta r}, \quad (13)$$

where $\mathbf{r} = (x, y, z)$ is the electron position and $r = |\mathbf{r}|$. Different values of θ define different trial wavefunctions and therefore different variational energies $E(\theta)$.

Electron positions were updated in all three spatial dimensions, and the resulting samples were used

in the Monte Carlo estimator for the variational energy.

For each sampled configuration $\mathbf{r}_i$, I evaluated the local energy using the Eq. (2) and the Laplacian was computed numerically using the finite-difference second-derivative routine validated previously, applied along each Cartesian direction. The variational energy for a given parameter value was then estimated using the Monte Carlo energy estimator of Eq. (5), with the local energy and probability density evaluated at $\psi(\mathbf{r}; \theta)$ after discarding a 20% burn-in segment to ensure that the walkers had equilibrated before sampling.

To optimise the variational parameter θ , I used simulated annealing in which each iteration proposes a random step in parameter space, $\theta' = \theta + \Delta\theta$, where the step $\Delta\theta$ was drawn from a uniform distribution over $[-\Delta\theta_{\max}, \Delta\theta_{\max}]$. This proposal was then accepted or rejected according to the probability

$$\alpha = \min\left(1, e^{-(E(\theta') - E(\theta))/T}\right), \quad (14)$$

where T is an effective temperature that decreases every n iterations, which allows occasional uphill moves and prevents the optimisation from getting trapped in local minima. Accepted steps update the parameter, $\theta \leftarrow \theta'$, and the energy evaluation is repeated. This procedure generates a random walk in parameter space biased towards lower-energy values of θ , and the point at which $E(\theta)$ ceases to decrease is taken as the variational optimum.

Because the VMC method requires many energy evaluations, the Metropolis sampler was optimised for efficiency using multiple walkers and NumPy vectorisation. Rather than advancing a single chain, an ensemble of walkers is evolved in parallel, allowing proposal steps, acceptance tests, and PDF evaluations to be performed on whole arrays. This substantially reduces the computational cost per sample

To tune the VMC method, the 3D Metropolis proposal step was adjusted to achieve a position acceptance rate of approximately 50%, and the sampler was run with 200 walkers to limit computational cost. For the parameter update, several proposal widths were tested and $\Delta\theta = 0.04$ was chosen, yielding a stable acceptance rate of about 40%. An effective temperature of $T = 0.05$ allowed occasional uphill moves, preventing stalling due to

statistical noise, and was reduced by a factor of 0.6 every ten iterations to refine the search near the minimum.

The accuracy of the optimised parameter can, in principle, be improved by reducing the θ step size so that the optimiser makes finer adjustments near the minimum. However, if the step size becomes too small, the resulting change in energy is comparable to the statistical fluctuations of the Monte Carlo estimator. In this noise-dominated regime, the Metropolis test cannot reliably distinguish whether a proposed update raises or lowers the true energy, causing a drop in acceptance rate and preventing further meaningful progress.

To resolve smaller energy differences, one would need to reduce the statistical noise of each energy evaluation, for example by increasing the number of walkers or the number of Metropolis iterations. This increases the cost per update, so in practice a balance must be struck between precision and computational expense. In this work, I used 1000 walkers for each energy evaluation, which provided a sufficiently low variance for stable optimisation.

Results: As Figure 3 shows, after approximately 150 updates the parameter stopped drifting and the energy no longer improved, giving an optimal value of $\theta = 0.987$ and an energy estimate of $E = -0.49998$, within 10^{-4} of the exact ground-state value.

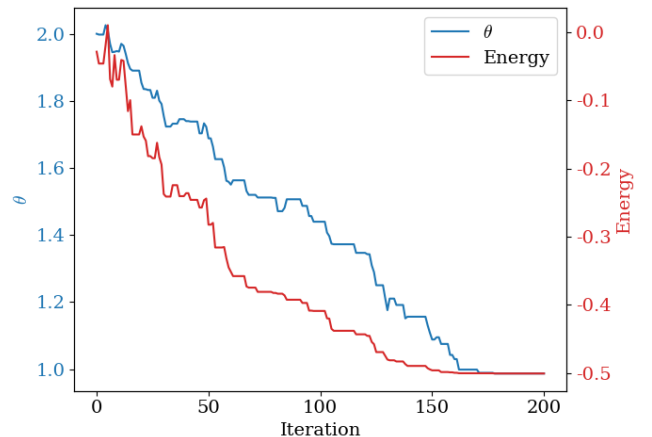


Fig. 3. Convergence of the variational parameter θ and the corresponding energy estimate during VMC.

D. Hydrogen Molecule in 3D

Method: The VMC method was then extended to the hydrogen molecule, for which

the wavefunction is non-separable and electron–electron repulsion must be included explicitly. The Born–Oppenheimer approximation was employed, with the nuclei fixed at positions $\mathbf{R}_A$ and $\mathbf{R}_B$ separated by an internuclear distance R .

The trial wavefunction was replaced by a two-electron correlated form,

$$\psi(\mathbf{r}_1, \mathbf{r}_2; \boldsymbol{\theta}) = \left[e^{-\theta_1(r_{1A}+r_{2B})} + e^{-\theta_1(r_{1B}+r_{2A})} \right] \times \exp \left[-\frac{\theta_2}{1 + \theta_3|\mathbf{r}_1 - \mathbf{r}_2|} \right]. \quad (15)$$

which depends on three variational parameters and incorporates both electron–nuclear attraction and explicit electron–electron correlation; however, it remains a restricted ansatz and does not represent the exact ground-state wavefunction.

Sampling was carried out using the same method as for the hydrogen atom, except that metropolis trial moves were now proposed independently for both electrons. Each step updated the six-dimensional configuration $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2)$ by adding independent three-dimensional uniform steps to $\mathbf{r}_1$ and $\mathbf{r}_2$.

The local energy also changed, now including two Laplacians and five Coulomb terms,

$$E_L = -\frac{1}{2} (\nabla_1^2 + \nabla_2^2) \psi / \psi - \frac{1}{r_{1A}} - \frac{1}{r_{1B}} - \frac{1}{r_{2A}} - \frac{1}{r_{2B}} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} + \frac{1}{R}. \quad (16)$$

with the Laplacians evaluated using the finite-difference scheme validated in the derivative-accuracy test. The same simulated-annealing scheme was used to optimise the three-dimensional parameter vector $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$.

Across all proton separations, a single set of numerical parameters was used, chosen through short preliminary tests to ensure stable sampling and reliable energy estimates. The Metropolis proposal step for the electron coordinates was fixed at $\Delta r = 1.1$ a.u., giving an acceptance rate of roughly 50–60%. Each Metropolis evaluation used 100 iterations with a 20% burn-in; during the θ -minimisation, the initial energy was computed with 1000 walkers, while proposed parameter sets were evaluated with 500 walkers, providing sufficiently low Monte Carlo noise without excessive computational cost. Parameter updates were proposed as $\theta'_i = \theta_i + 0.3u_i$ with $u_i \in [-1, 1]$, a step size that

produced resolvable energy changes while keeping proposals within physically sensible bounds. Simulated annealing was performed with an initial temperature $T_0 = 0.5$, halved after every five accepted updates to allow broad exploration early on and refinement near the minimum. For each proton separation, 100 annealing iterations were enough for both the parameters and the energy estimate to stabilise, after which a higher-statistics Metropolis run (1000 walkers, 100 iterations) was used to obtain the final energy values for the Morse fit.

Results: To study how the ground-state energy of the hydrogen molecule varies with internuclear separation, I performed VMC calculations for bond lengths in the range $0.2 \leq r \leq 5.0$ a.u.. For each r , the Metropolis sampler generated configurations in the six-dimensional electronic space, and the corresponding local energies were averaged to estimate $E(r)$.

I fitted the resulting energy curve to a Morse potential as can be seen in Fig. 4,

$$V(r) = E_{\min} + D(1 - e^{-a(r-r_0)})^2. \quad (17)$$

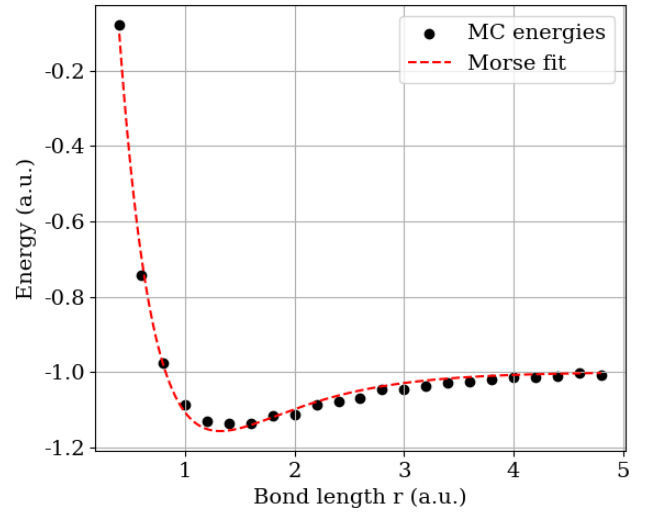


Fig. 4. Monte Carlo energy estimates for H_2 as a function of bond length, together with a Morse potential fit. The minimum of the fitted curve gives the predicted equilibrium separation. Monte Carlo statistical error bars (± 1 s.e.) are smaller than the marker size and therefore not visible.

This yielded the parameter values $D = 0.157$ a.u., $a = 1.388$ a.u. $^{-1}$, $r_0 = 1.322$ a.u., $E_{\min} = -1.157$ a.u. The fitted value of r_0 provides the predicted equilibrium bond length, while $E_{\min}$ is the corresponding minimum energy.

In addition to the energy calculations, I visualised the electron probability density. The resulting two-dimensional histogram is shown in Fig. 5.

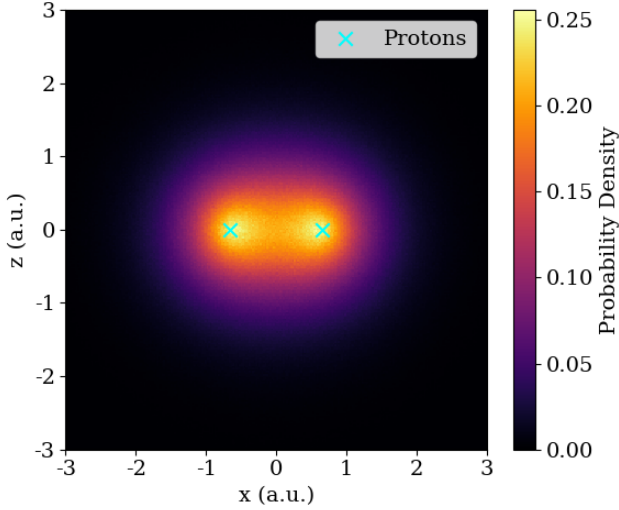


Fig. 5. Two-dimensional histogram of sampled electron positions for H_2 , projected onto the x - z plane. The density is highest between the nuclei, characteristic of a bonding molecular orbital.

The fitted parameters are close to the known values for H_2 , though small discrepancies remain: the equilibrium separation is slightly below the experimental value of 1.40 a.u. [5], and the dissociation energy D is marginally lower than the experimental 0.17 a.u. The predicted equilibrium separation differs from the experimental value by approximately 5.6%, while the dissociation energy is underestimated by about 8%.

Statistical uncertainties on the variational energy estimates were obtained using block averaging of the local energy samples to account for correlations introduced by the Metropolis algorithm. The sampled local energies were divided into N_b blocks, and the uncertainty on the mean energy was estimated from the variance of the block means[6],

$$\sigma_E = \frac{\sigma_{\text{block}}}{\sqrt{N_b}}, \quad (18)$$

where σ_{block} is the standard deviation of the block-averaged energies. Using this approach, the optimised ground-state energy of the hydrogen molecule was estimated as $E = -1.157 \pm 0.001$ a.u. The deviation of this value from the accepted ground-state energy is an order of magnitude larger than the Monte Carlo statistical uncertainty, indicating that the discrepancy is dominated by systematic error

arising from the restricted variational ansatz, rather than finite sampling noise.

IV. CONCLUSION

In this project I implemented Variational Monte Carlo to estimate the ground-state energy of the hydrogen molecule and quantified the deviation of the calculated results from accepted experimental values. The numerical components of the method were first validated on systems with known analytical solutions: the finite-difference Laplacian was shown to operate reliably over an appropriate range of step sizes, and the Metropolis sampler reproduced the energies of the one-dimensional and three-dimensional hydrogen problems to high precision. Extending the method to H_2 , I performed Metropolis sampling in the six-dimensional electronic space and optimised the variational parameters using simulated annealing. This resulted in a Morse potential fit yielding an equilibrium separation of $r_0 = 1.3223$ a.u. and a minimum energy of $E_{\text{min}} = -1.1565$ a.u., corresponding to deviations of approximately 5.6% in bond length and 8% in dissociation energy relative to accepted experimental values, with the dominant source of error arising from the restricted ansatz.

Overall, the results demonstrate that efficient Monte Carlo sampling provides a quantitatively reasonable description of H_2 and successfully captures its essential physical features.

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